1.06 Quadratics: Introduction

Chapter 2 · Original slide 5 · 1: Solving Quadratic Equations

$x^2 + 5x = 6$

There are three ways of solving a quadratic equation.

One is by *completing the square*, which we'll do later.

What are the other two ways?

By factorisation

Using the quadratic formula.

If $ax^2 + bx + c = 0$ then

$x^2 + 5x - 6 = 0$

Put in form.

$ax^2 + bx + c = 0$ with 0 on one side

Factorise. If the product of two things is 0, at least one must be 0.

Chapter 2 · Original slide 6 · 1: Solving Quadratic Equations

Solving without factorising

If the subject only appears once however, it might be easier not to expand out/factorise:

If you can't see why the $\pm$ is required, think about the solutions to: $x^2 = 4$.

$2^2 = 4$, but $(-2)^2 = 4$ as well!

So $x = \pm 2$.

Quadratics 'in disguise'

When we have an expression like say $x^2 + 3x - 2$, we say it is "quadratic in x". In trigonometry you will have to solve equations like $(\sin x)^2 + 3\sin x + 2$. We say that the expression is "quadratic in $\sin x$".

Either use a suitable substitution so that you have a 'normal' quadratic, or go straight for the factorisation if you're feeling more confident (recommended!).

Solve $x - 6\sqrt{x} + 8 = 0$

substitution

hardcore

1.06 Quadratics: Introduction

Chapter 2 · Original slide 7 · Test Your Understanding

1
Solve $(x+3)^2 = x+5$ using factorisation.

2
Solve $(2x+1)^2 = 5$

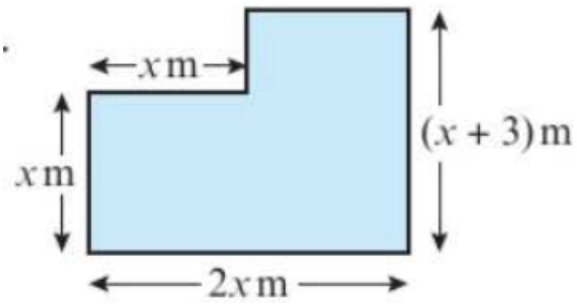
3
Solve $\sqrt{x+3} = x-3$

4
Solve $2x + \sqrt{x} - 1 = 0$

"Use of Technology":

The 'equation' mode on your calculator will solve quadratics in the form $ax^2 + bx + c = 0$. When you're asked for the 'order', use 2 (we'll see why later).

Chapter 2 · Original slide 8 · Exercise 2A/2B Pure Y1 Pages 20-22



3 Solve the following equations:

i $6x^2 - 7 = 11x$

h $5x^2 - 10x^2 = -7 + x + x^2$

4 This shape has an area of $44\,\text{m}^2$. Find the value of x.

5 Solve the equation $5x + 3 = \sqrt{3x+7}$.

1.06 Quadratics: Introduction

Chapter 2 · Original slide 9 · Exercise 2A/2B Pure Y1 Pages 20-22

2 Solve the following equations using the quadratic formula. Give your answers to three significant figures.

h $2x^2 + 11x = 5x^2 - 18$

3 For each of the equations below, choose a suitable method and find all of the solutions. Where necessary, give your answers to three significant figures.

e $(2x + 8)^2 = 100$ **f** $6x^2 + 6 = 12x$

Chapter 2 · Original slide 10 · Exercise 2A/2B Pure Y1 Pages 20-22

1

Extension: (Full Database: http://www.drfrostmaths.com/resources/resource.php?rid=268)

1.06 Quadratics: Introduction

Chapter 2 · Original slide 11 · 2: Completing the Square

“Completing the square” means putting a quadratic in the form $(x+a)^2+b$

or $a(x+b)^2+c$

The underlying reason we do this is because $\mathbf{x}$ only appears once in the expression (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

b. Sketching Quadratics

If we have a completed square:

$(x+4)^2-7=0$

we saw at the start of the chapter how we could rearrange to make x the subject.

Indeed using the quadratic formula is actually solving the quadratic by completing the square – it’s just someone has done the work for us already!

We’ll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

c. In integration

In Further Maths, completing the square allows us to ‘integrate’ expressions like:

$\int \frac{1}{x^2-4x+5}dx$

(you will cover integration later this module)

Chapter 2 · Original slide 12 · Completing the Square Recap

Textbook Note:

The textbook uses the formula

$x^2+bx=\left(x+\frac{b}{2}\right)^2-\left(\frac{b}{2}\right)^2$

and similarly

$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+\left(c-\frac{b^2}{4a}\right)$

My personal judgement is that it’s not worth memorising these and you should instead think through the steps. Even the textbook agrees!

Expand:

$(x+\mathbf{9})^2=$
$(x-\mathbf{5})^2=$

What do you notice about the relationship between the bold numbers?

Therefore if we had x^2+12x, how could we write it in the form $(x+a)^2+b$?

Since $(x+6)^2=x^2+12x+36$, we want to discard the 36, so ‘throw it away’ by subtracting.

Further Examples:

$x^2+8x=$
$x^2-2x=$
$x^2-6x+7=$

Notice that despite the a being negative, we still subtract after the bracket as $(-1)^2$ is positive.

1.06 Quadratics: Introduction

Chapter 2 · Original slide 13 · Completing the Square

Express $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x + c)^2$

It may help to write in the form $ax^2 + bx + c$ first.

Chapter 2 · Original slide 14 · Test Your Understanding

Express $3x^2 - 18x + 4$ in the form $a(x + b)^2 + c$

Express $20x - 5x^2 + 3$ in the form $a - b(x + c)^2$

Chapter 2 · Original slide 15 · Solving by Completing the Square

Solve the equation:

$3x^2 - 18x + 4 = 0$

Reflections: Previously we factorised out the 3. This is because $3x^2 - 18x + 4$ on its own is an expression, so dividing by 3 (instead of factorising) would change the expression.

However, in an equation, we can divide both sides by 3 without affecting the solutions.

Chapter 2 · Original slide 16 · Exercise 2C/2D Pure Y1 Page 23-24

1 Complete the square for the expressions:

a $x^2 + 4x$
b $x^2 - 6x$
c $x^2 - 16x$
d $x^2 + x$
e $x^2 - 14$

2 Complete the square for the expressions:

a $2x^2 + 16x$
b $3x^2 - 24x$
c $5x^2 + 20x$
d $2x^2 - 5x$
e $8x - 2x^2$

1.06 Quadratics: Introduction

Chapter 2 · Original slide 17 · Exercise 2C/2D Pure Y1 Page 23-24

4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\text{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \text{and} B$ are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

Chapter 2 · Original slide 18 · Proving the Quadratic Formula

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Just use exactly the same method as you usually would!